

CiviCite: A Verified, Citation-First Retrieval-Augmented Assistant for Swedish Public Services

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Abstract

People navigating Swedish public services, especially newcomers, need answers that are correct, current and traceable to an official source. General-purpose chatbots fail in exactly the ways that matter here: they invent deadlines, amounts and rules. CiviCite is a retrieval-augmented assistant built around three guarantees. Every factual sentence must cite a retrieved source. A sentence-level verifier checks term overlap and *number consistency* against the cited text. The system abstains when the evidence is weak. Date arithmetic, a frequent source of errors, is delegated to a deterministic tool that implements the Swedish calendar rules. Personal identity numbers and other personal data are redacted before any text reaches a model or a log, and retrieved passages that attempt prompt injection are discarded. On a 34-question development set, the LLM-free extractive baseline reaches 1.00 retrieval recall@5, 0.846 answer accuracy, 0.857 abstention F1 and 1.00 faithfulness, with sub-millisecond latency. An evaluation harness gates these numbers in continuous integration and runs the same suite against any OpenAI-compatible model.

1 Introduction

Retrieval-augmented generation (RAG) [1] grounds language models in external documents, but grounding alone does not guarantee faithfulness: models still mix retrieved facts with parametric memory, cite the wrong passage, or answer questions the documents do not cover [2, 3]. In public-service guidance the cost of such errors is concrete: a wrong deadline for reporting a move (*flyttanmälan*) or for applying for temporary parental benefit (VAB) can mean fees or lost benefits.

CiviCite treats *verifiability* as the primary design goal rather than fluency:

- **Citation-first answers:** every claim carries a source number, and the UI shows the cited text.
- **Post-hoc verification:** each sentence is checked against its cited source, and unsupported sentences are flagged (or, in strict mode, the answer is replaced by an abstention).
- **Calibrated abstention:** an evidence score decides when to answer at all.
- **Deterministic tools** for date arithmetic, following the Swedish rule that statutory deadlines falling on a Saturday, Sunday, public holiday, Midsummer Eve, Christmas Eve or New Year’s Eve move to the next working day [9].
- **Privacy and robustness:** Luhn-validated redaction of *personnummer* and *samordningsnummer*, e-mail, phone, IBAN and card numbers, and an indirect-prompt-injection filter for retrieved text [5].

2 System design

Ingestion. Markdown, HTML (navigation, scripts and footers removed, headings preserved) and PDF documents are split into heading-aware chunks that never span two sections. Each chunk keeps its title, section, agency and URL. A polite crawler (robots.txt, rate limiting, prefix-restricted, de-duplicated by content hash) converts agency web pages into this format.

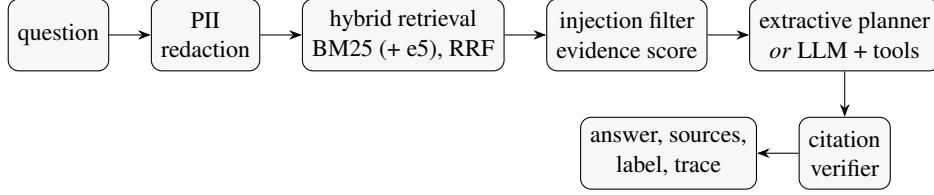


Figure 1: CiviCite pipeline.

Retrieval. BM25 [6] over stop-word-filtered, lightly stemmed English/Swedish terms is fused with optional multilingual E5 embeddings [8] by reciprocal rank fusion [7]. A small Swedish→English glossary expands queries when the corpus language differs from the question language.

Evidence score and abstention. For query terms T (each with its glossary alternatives) and a chunk c , the IDF-weighted coverage is

$$\text{cov}(c) = \frac{\sum_{t \in T} w_t \cdot \mathbf{1}[t \in c]}{\sum_{t \in T} w_t}, \quad w_t = \begin{cases} \text{idf}(t) & t \text{ in collection} \\ \max \text{idf} & \text{otherwise.} \end{cases}$$

Assigning maximal weight to out-of-collection terms makes questions about uncovered topics (“opening hours in Malmö”) score low even when generic words match. CiviCite abstains when the best coverage among the top three chunks is below a threshold τ (default 0.34, chosen by the sweep in Table 2).

Answering. In *extractive mode* (no LLM) sentences from the top chunks are scored by IDF-weighted overlap, down-weighting words that merely repeat the document title, boosting sections whose heading matches the question type (how/who/deadline/how many), and weighting by chunk evidence. A rule-based planner detects deadline questions, extracts the rule (“within one week”, “no later than 90 days”, “no later than 2 May”) from the cited sentence and calls the deadline tool. In *LLM mode* the model receives the numbered sources and three tools (`search_documents`, `calculate_deadline`, `next_business_day`) in an OpenAI-compatible tool-calling loop. The system prompt forbids outside knowledge and own date arithmetic, and the agent falls back to extractive mode if the model is unreachable.

Verification. Answers are split into sentences, and citation markers placed after the full stop are normalised. A sentence is a *claim* if it has at least three content terms and is not a meta statement. A claim is supported if (i) it cites a valid source, (ii) at least 50 % of its content terms occur in the cited chunk (title and section included), and (iii) every number in it occurs in the cited chunk. For sentences marked `[calc]`, date words and numbers must match the recorded tool results, and the remaining words must match the cited rule. Faithfulness is the share of supported claims.

3 Evaluation

The development set has 34 questions: 26 answerable (24 English, 2 Swedish; 3 require date computation) and 8 unanswerable. The unanswerable questions include out-of-domain questions, an instruction-injection attempt, and two *hard negatives* that share vocabulary with the corpus (“interest rate on CSN loans”, “work permit”). Metrics: retrieval recall@5 and MRR at document level; answer accuracy (required phrases present, abstentions count as wrong); abstention precision/recall/F1 with “unanswerable” as the positive class; faithfulness; citation precision; tool/date accuracy; latency.

Error analysis. All four wrong answers retrieved the correct document (recall is perfect) but selected the wrong sentence. For example, “How do I apply for an identity card?” returned the eligibility sentence instead of the procedure, and a Swedish question matched the generic “deadline” sentence of another

Metric (extractive mode, BM25 only)	Value
Retrieval recall@5 / MRR	1.000 / 1.000
Answer accuracy (26 answerable)	0.846
Abstention precision / recall / F1	1.000 / 0.750 [†] / 0.857
Faithfulness (verified claims)	1.000
Citation precision	0.886
Deadline tool accuracy	1.000 (3/3)
Latency p50 / p95	0.8 ms / 1.1 ms

Table 1: Development-set results, 2026-10-05. [†]6 of 8 unanswerable questions abstained; no answerable question was wrongly abstained.

τ	0.10	0.20	0.25	0.30	0.34	0.40	0.50	0.60
Abstention F1	0.400	0.667	0.769	0.857	0.857	0.875	0.842	0.727
Answer accuracy	0.846	0.846	0.846	0.846	0.846	0.808	0.769	0.654

Table 2: Abstention-threshold sweep. Higher thresholds trade answer coverage for safer abstention.

document. Both remaining false answers are the hard negatives. Lexical evidence cannot tell that “interest rate on CSN loans” is not covered by a chunk that discusses CSN loans. These are the cases where LLM mode with semantic judgement and dense retrieval is expected to help. The harness measures this directly with `civicate eval -mode llm`. Because the heuristics were tuned on this set, the numbers are development-set results. A held-out test set is required before claiming generalisation.

Safety tests. Unit tests confirm that (i) identity numbers never appear in traces, (ii) a chunk containing “ignore all previous instructions” is excluded from the context and reported, (iii) an LLM answer with an invented deadline and fee is labelled *unverified* and replaced by an abstention in strict mode, and (iv) an unreachable model triggers a transparent fallback.

4 Limitations and ethics

The bundled corpus consists of short demonstration texts, not official guidance. Real use requires ingesting the agencies’ pages and re-indexing as rules change. Lexical verification catches invented numbers and off-topic sentences but not every paraphrased misstatement; natural-language-inference verification and self-critique [4] are natural extensions. Questions may contain personal data, so redaction happens before retrieval, generation and logging, in line with GDPR data-minimisation. The system is an information aid, not legal advice, and always links to its sources.

5 Conclusion

Making verification, abstention and deterministic tools first-class components yields an assistant whose answers can be checked sentence by sentence. A deterministic baseline already achieves perfect faithfulness on the development set, and the remaining errors are measurable targets for LLM-based improvements.

References

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